Vertical Integration and the "Sourcing Style"*

Jiaxu Zhang †

January, 2026

Abstract

I build a model of the boundary of the firm with asset ownership to show how firms can complement their organizational form by choosing different sourcing systems. I show that, given that both the "Japanese" style sourcing and the "American" style sourcing ensure similar quality levels, vertical integration and the "Japanese" style sourcing are complementary. Sourcing from related parties is more likely to result in a long-term relationship with high-frequency shipments and small shipment sizes. Complementarity is stronger when the asset of the upstream firm has greater value. Combining the ORBIS data, which has large-scale firm-to-firm ownership information, with the Panjiva data, which includes all the maritime firm-to-firm transaction records, I provide empirical evidence consistent with the above theoretical predictions.

Keywords: supply chain; vertical integration; global sourcing; incentive

JEL Codes: F1, L14, L22

*I thank Matthias Fahn, Chang Sun, Bingjing Li, and Jin Li for their helpful comments.
†Jiaxu Zhang: HKU Business School, University of Hong Kong, zhjiaxu@connect.hku.hk.

1 Introduction

In recent decades, global supply chains have become increasingly important in international trade. Downstream firms source inputs from upstream firms through various contractual forms. For example, in the automotive industry, American automakers like General Motors typically select the lowest bidder each year. At the same time, Japanese automobile firms, such as Toyota, sign long-term contracts with high-performing suppliers and frequently conduct just-in-time transactions with them. Firms typically choose between American-style (A style) sourcing—large, infrequent shipments with mandatory inspections—and Japanese-style (J style) sourcing—small, frequent shipments, and long-term relationships. The A style relies on spot contracts and inspection for quality control, while the J style uses relational contracts and repeated interactions to incentivize quality. (Taylor and Wiggins, 1997) Recent studies show a shift towards J style sourcing as trade environments stabilize and competition from low-income countries rises. (Heise, Pierce, Schaur and Schott, 2024; Fernandes and Tang, 2020; Ortiz, 2022) This paper studies how the complementarity between a firm's ownership structure and its purchasing behavior enables a firm to switch from A style to J style.

This paper combines the partial equilibrium trade model of Heise et al. (2024)[1] with the idea of asset ownership in the theory of firm boundaries formalized by Gibbons (2005). By allowing either the downstream or the upstream firm to own the assets, that is, to have the right to control the products produced by the assets and the right to resell the assets, buyers can make a tradeoff between the benefits and losses of vertical integration of suppliers when choosing a purchasing system, and thus use appropriate purchasing systems to assist ownership structures.

In my model, upstream firms owning the asset represent the case of non-integration because the asset belongs to producers. Naturally, vertical integration means that downstream firms own the asset. The party that owns the asset has the right to sell it and receive a one-time benefit. From this, we can see that when the downstream firm does not acquire the upstream firm, that is, when the assets are owned by the upstream firm, the upstream firm has a greater outside option. Therefore, the downstream firm needs to put more effort into giving the upstream firm an incentive to produce high-quality products.

Considering that buyers who choose A style only provide incentives to suppliers through inspections of each shipment, and they use spot contracts, whether the supplier has assets will not affect the buyer's choice of the lowest-cost seller in the entire market, as long as

[1] Heise et al. (2024) extends the model of Taylor and Wiggins (1997) by adding a parameter ρ to the discount rate to measure policy uncertainty and forcing buyers who choose A style to monitor each transaction with 100% probability.

the monitoring cost is exogenously given. However, buyers who choose the J style need to provide incentives to suppliers, on the condition that the net present value of long-term contracts covers the benefits the upstream firm can obtain from a one-time default. This means suppliers with assets need higher premiums to be incentivized. By formally modeling the aforementioned situation, my model shows the complementarity between buyers' vertical integration and the choice of J style. In addition, I also show that the complementarity between vertical integration and J style sourcing is stronger when the asset of the upstream firm has a higher value. My theory gives firms, especially multinational firms, greater flexibility in choosing the structure of global supply chains and makes supply chains less susceptible to negative shocks.

Next, I combine transaction-level US import data provided by Panjiva with global ownership data provided by the ORBIS database[2] to identify trade patterns between pairs of firms and which transactions occur between related parties. I use the cleaned firm names and their countries as keys to match the two datasets, and treat transactions in Panjiva data that occur between firm-firm combinations that can be matched to firm-firm pairs with ownership relationships in ORBIS data as transactions between related parties. I construct a dummy variable to test the relationship between trade patterns and the presence of vertical integration between downstream firms and upstream firms. A shortcoming is that this study lacks a satisfactory micro-level exogenous shock to identify a causal relationship between sourcing patterns and vertical integration, but making good use of the detailed transaction-level data provided by the Panjiva allows me to construct variables that fully fit the model and control for different combinations of fixed effects to demonstrate a relationship that is robust across multiple specifications.

My empirical results confirm the model's predictions: firms' sourcing style is more likely to be J style when they integrate their suppliers. I find that related parties trade more frequently than non-related parties, and that the average quantity per shipment for related parties is smaller than for non-related parties. I also find that the complementarity between vertical integration and J style sourcing is stronger when the asset of the upstream firm has a higher value. Specifically, if the asset value of the subsidiary controlled by the owner exceeds the 75th percentile, the frequency of shipping will be further increased by 20%-40% compared with the counterparty with a smaller subsidiary.

This paper makes contributions to three strands of literature. First, my study extends the research on firms' adoption of different procurement styles by adding the interaction between purchasing systems and organizational structures. As Milgrom and Roberts (1990)

[2]I process the ORBIS data following the method suggested by Kalemli-Ozcan, SSrensen, Villegas-Sanchez, Volosovych and Yesiltas (2015)

proposed, when analyzing firms' strategies, all elements should be considered together, as the firm's various strategies are complementary. It is not uncommon for firms to directly acquire suppliers when importing inputs, indicating that a firm's procurement strategy and its choice of organizational form interact. After Taylor and Wiggins (1997) first use the partial equilibrium model to characterize American and Japanese style sourcing, few studies have discussed in depth the relationship between sourcing systems and organization forms. Heise et al. (2024) extends the model by adding an exogenous shock to examine how sourcing style changes with policy uncertainty. They focus on how exogenous policies affect the choice of procurement system for arm's-length trade. However, I focus on the complementarity between firms' integration choices and procurement system choices, providing a more micro perspective in this paper.

Second, it helps us better understand how vertical integration shapes global supply chains. As Fort (2023) puts forward, the in-house production outside the domestic country is becoming more popular. In the literature on the relationship between vertical integration and global supply chains, existing research provides us with multiple ways to optimize supply chains through vertical integration. Antràs and Helpman (2006) believes that vertical integration can reduce contractual friction and information asymmetry, and that the position of information asymmetry in the supply chain also matters. They point out that it is better to do vertical integration when the contractibility of an input is provided by the supplier. Based on the property rights idea in Antràs and Helpman (2006), Antràs and Chor (2013) and Alfaro, Antràs, Chor and Conconi (2019) add sequential production and show that whether a firm integrates its upstream or downstream partner depends on the substitutability or complementarity of inputs throughout the supply chain and the elasticity of demand for its final product. Different from the property rights theory, Forbes and Lederman (2009) and Costinot, Oldenski and Rauch (2011) argue that integrating suppliers actually enhances the firm's ex-post adaptation capabilities and thus improves the firm's performance. They both provide simple models grounded in adaptation theory and supported by empirical evidence. In addition, Berlingieri, Pisch and Steinwender (2021) argues that whether to integrate or not depends on the technological importance of inputs, whereas Grossman and Helpman (2004) shows that the make-or-buy decision depends on productivity and the ability to monitor the headquarters. Although these studies provide a wealth of information on how vertical integration promotes global supply chains, they all focus on studying how the structure of the supply chain affects the firm's choice of when to conduct vertical integration endogenously. However, my study mainly shows how the firm's organizational form interacts with the supply chain when vertical integration is exogenously given.

Third, this paper makes both theoretical and empirical contributions to the literature

on relational contracts within firms. Baker, Gibbons and Murphy (2002) builds a detailed theoretical model to show the four organizational forms [3] that firms will choose when faced with whether to use relational contracts and whether to do vertical integration. They explain that the key differences in relational contracts, both between and within firms, are mainly reflected in the parties' motivations to default. My paper combines the models of these four situations with global sourcing and comfirms that vertical integration between firms can strengthen the relational contract. I also empirically demonstrate this conclusion using detailed firm-level trade data and foreign ownership data, providing empirical evidence for past research.

The paper proceeds as follows. In Section 2, I describe the data and construct the key variables in the empirical analysis. I convey the motivating facts in Section 3. Section 4 introduces the theoretical model to rationalize the motivating facts and provides the main argument and several comparative statics. I introduce the empirical analysis for testing the additional predictions in Section 5. I conclude in Section 6.

2 Data

In this section, I introduce the datasets I use, which primarily consist of trade data and ownership data. I describe how to match these two datasets to identify trade between related parties. I also describe the construction of the main variables used in the empirical analysis.

2.1 Firm-to-firm Trade Data

I use the US Bills of Lading data from 2007 to 2020, provided by Panjiva, which contains all maritime import transactions of US companies. Panjiva provides the names and unique IDs of the consignees (the US firm) and the shippers (the foreign firm) for each shipment record, the addresses of both parties, the arrival date of each shipment, the Harmonized System (HS) product code in six digits, the weight of the product in kilograms,[4] and the total quantity of each shipment,[5] etc. Panjiva data provides rich information on firm-to-firm trade, enabling us to conduct micro-level analysis.

[3] Baker et al. (2002) divides potential organizational form and contractual form into four categories: spot outsourcing, spot employment, relational outsourcing, and relational employment. They use "outsourcing" refer to non-integration, "employment" refer to integration, "spot" refer to spot contract, and "relational" refer to relational contract. The details are in their Figure 1.

[4] Panjiva provides both tons and kilograms as units for shipment records, but we use kilograms in our analysis to prevent rounding errors.

[5] The quantities in the raw data have units such as CTN (carton/cartons), PCS (piece/pieces), PKG (package/packages), etc. Different units may be used for the same product in different shipment records.

2.2 Ownership Data

I use ORBIS, the global firm-level database maintained by Bureau van Dijk (BvD), to construct all the potential cross-country related party pairs. The controlling shareholder (csh) files from this database provide all the levels of controlling shareholders and the end subsidiary. Following the definition for US firms of foreign multinational enterprises (MNEs) and US MNEs in Antràs, Fadeev, Fort and Tintelnot (2022), I identify foreign MNEs and US MNEs using csh files from 2011 to 2017.[6]

The csh files of ORBIS data include all the controlling shareholder-subsidiary pairs, the total ownership and direct ownership percentages between them, and the csh level of each controlling shareholder. Using this structure, I can identify all the business groups: the subsidiary in the csh data is the one at the end of each "ownership tree". The ultimate owner for each "ownership tree" is the one with the highest csh level. Thus, I can identify foreign and US MNEs based on the location of their ultimate owners. The details and examples can be seen in Appendix B.1.

After identifying all the business groups with foreign MNEs or US MNEs, I define cross-country related parties for US firms as the pair of any US firm-foreign firm combinations within the same business groups.

2.3 Matching Panjiva Data and ORBIS Data

To investigate whether related parties exhibit different patterns in global sourcing, I need to combine firm-to-firm trade data with ownership relationships.

The firms in the Panjiva and ORBIS datasets do not have a unified identifier, so I use the firm's name and country to match the two datasets. To be consistent with the ORBIS data, I only use trade data from 2011 to 2017. I first clean the firm names in both datasets and perform exact matches. As long as the cleaned firm names and the countries of firms from Panjiva data and ORBIS data, respectively, are the same, I consider them to be the same firm. The matching process is described in Appendix B.2 in detail.

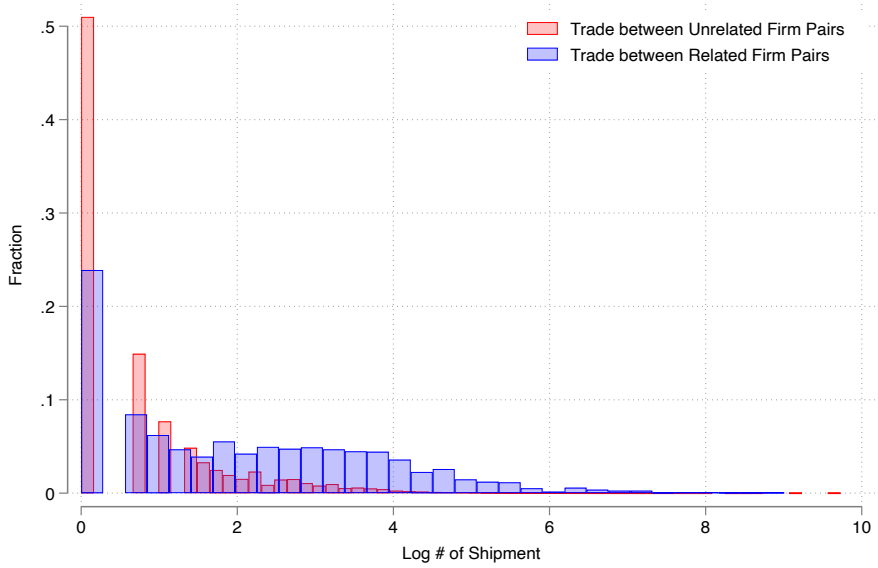
To assess how much attrition this matching approach will generate, I compare descriptive statistics of the matched samples with those reported in Antràs et al. (2022). Using this method to match the Panjiva-ORBIS data, a similar number of MNEs and a similar share of related-party trade are identified.

[6] Antràs et al. (2022) divides firms that are not purely US domestic firms into foreign MNEs and US MNEs. Foreign MNEs are US firms that are majority-owned by foreign firms, while US MNEs are US firms that have a majority-owned foreign affiliation and are majority-owned by US firms. They use the firm's ultimate owners to make the designation.

3 Motivating Facts

In this section, I present the motivating facts about the relationship between sourcing patterns and related party trade using matched Panjiva-ORBIS data. Figure 1 shows the distribution of the shipping frequency in 2011 for related party trade and non-related party trade. It can be seen that related party trade has a higher frequency of shipments than non-related party trade on average. Among non-related parties, over 50% only made one transaction in 2011, a much smaller percentage for related parties. Furthermore, for related parties, the ratio of 2-6 transactions per year is significantly higher. This fact suggests that related parties are more likely to choose J style sourcing, which is characterized by higher shipping frequency.

Figure 1: Distribution of Log # Shipment, 2011



I further illustrate this phenomenon using a regression that controls for firm size and strict fixed effects. Using higher shipping frequency with smaller quantity per shipment as the measurement of the preference of J style, the regression equation is as follows:

$$Y_{ijt} = \beta_1 \mathbb{1}\left\{\text{Related Party Trade}_{ijt}\right\} + \beta_2 \log(TotQuantity_{ijt}) + \delta_{it} + \delta_{jt} + \delta_{it'} + \delta_{jt'} + \epsilon_{ijt} \quad (1)$$

where t is the year from 2011 to 2017, Y_{ijt} can be both the number of shipments and the quantity per shipment. Variable of interests is $\mathbb{1}\left\{\text{Related Party Trade}_{ijt}\right\}$, this is an indicator equal to one if two firms are identified as related parties using the method mentioned

in Section 2. I control for firm size by adding the total quantity of trade $\log(TotQuantity_{ijt})$ between two firms to control for the fact that larger firms always trade more. δ_{it} is the importer-year fixed effect and δ_{jt} is the exporter-year fixed effect. $\delta_{it'}$ is the importer-first-transaction-month fixed effect and $\delta_{jt'}$ is the exporter-first-transaction-month fixed effect. I control for the month of the first transaction to ensure that each firm pair is likely to be involved in transactions for the same length of time within a fixed time period - this is because partners that start trading for the first time in December of a certain year are likely to have fewer transactions than partners that start trading in January, even if their trading frequency is high. Controlling time in this way can avoid the bias caused by different starting times of trade. ϵ_{ijt} is the error term.

It should be noted that I cannot control for the importer-exporter fixed effect because the related party status has little variation over time. If the importer-exporter fixed effect is controlled, there will be very few cases of integration vs. non-integration that can be identified, and most importer-exporter pairs will be treated as singletons, resulting in insufficient changes in the independent variables to identify statistically significant effects. Therefore, I choose to compare the sourcing patterns between related parties and non-related parties in a cross-section without pair fixed effect. Instead, I add δ_{it} and δ_{jt} to capture the time-varying characteristics of each trading partner. In addition, considering that the effects of integration may have a lag effect, and the related party status between firms will not vary frequently over time, Y_{ijt} and $\log(TotQuantity_{ijt})$ are both taken average using the moving average method to include the current year and the next two years.

I show the regression results in Table 1. The first two columns show the results of the number of shipments, and the last four columns show the results of the quantity per shipment. I directly take the moving average of quantity per shipment in the corresponding years in columns (3) and (4), while using the moving average of quantity and shipment, respectively, to calculate the quantity per shipment in columns (5) and (6).

With a loose fixed effect, column (1) shows that related parties have 0.166 log points more shipments than non-related parties. The positive effect is still significant when controlling for stronger fixed effects, importer-first-transaction-month, and exporter-first-transaction-month in column (2). After controlling for the time of the first transaction between firms, related parties have even more shipments than non-related parties. This result shows that related parties are more inclined to choose J style sourcing.

On the contrary, the last four columns show that sourcing activities between two related firms have lower transaction volumes. Among them, columns (3)-(4) calculate quantity per shipment first and then do the moving average, while columns (5)-(6) are the ratio of moving average quantity to moving average shipment. Columns (3)-(4) are closer to the average of

the lagged response of the current year and the next two years in definition, while columns (5)-(6) are simply the ratio of moving average quantity to moving average shipment. The coefficients in columns (5)-(6) will be the opposite value of those in columns (1)-(2) by variable construction. I will use the variable defined in the columns (3)-(4) later. Both sets of results demonstrate that, given the same trade volume between each other, related parties have less trade volume per transaction than independent firms.

Table 1: Relationship between Shipment Pattern and Related Party

Dep. Var.	$\log(Avg\ \#\ Shipment_{ijt})$		$\log(Avg\ \frac{quantity_{ijt}}{shipment_{ijt}})$		$\log(\frac{Avg\ quantity_{ijt}}{Avg\ shipment_{ijt}})$	
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\}$	0.166^a	0.264^a	-0.169^a	-0.268^a	-0.166^a	-0.264^a
	(0.010)	(0.031)	(0.010)	(0.031)	(0.010)	(0.031)
$\log(TotQuantity_{ijt})$	0.461^a	0.511^a	0.536^a	0.488^a	0.539^a	0.489^a
	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
Importer FE	Y	N	Y	N	Y	N
Exporter FE	Y	N	Y	N	Y	N
Year FE	Y	N	Y	N	Y	N
1st Transaction Mon. FE	Y	N	Y	N	Y	N
Importer-Year FE	N	Y	N	Y	N	Y
Exporter Year FE	N	Y	N	Y	N	Y
Importer-1st Transaction Mon. FE	N	Y	N	Y	N	Y
Exporter-1st Transaction Mon. FE	N	Y	N	Y	N	Y
Observations	5562091	2516789	5562091	2516789	5562091	2516789
# of Firm Pairs	2906559	1359826	2906559	1359826	2906559	1359826
R-squared	0.826	0.900	0.928	0.950	0.929	0.951

Notes: Standard errors are clustered at the firm-pair level. Moving averages are used for the dependent variables and the control variable $\log(TotQuantity_{ijt})$. The dependent variables in columns (3)-(6) are all quantities per shipment, but the third and fourth columns directly take the moving average of quantity per shipment in the corresponding years, while the fifth and sixth columns use the moving average of quantity and shipment, respectively, to calculate the quantity per shipment. The results of the two methods of calculating quantity per shipment are consistent with each other. Significance levels: a: 0.01, b: 0.05, c: 0.10.

4 Theoretical Model

In this section, I build a repeated game model with asset ownership and with global sourcing using different systems to rationalize the empirical findings. My model combines the asset ownership idea and the repeated game model introduced by Taylor and Wiggins (1997) and extended by Heise et al. (2024).

4.1 Sourcing Style with Asset Ownership

There is a country with a continuum of homogeneous sellers who can produce the same goods. These sellers need to produce products of quantity x for each shipment. Assuming that an asset and labor are the only inputs required to produce the products. The resale value of the asset is v. The alternative use value of the product is 0. ω is the productivity of the seller; I normalize the productivity to one without loss of generality. θ is the quality of the product. The output for each shipment is $x = \frac{1}{\theta}l$. There are two types of quality: high quality and low quality, $\theta \in \{\bar{\theta}, \underline{\theta}\}$. The cost of producing low-quality products is lower, so sellers may produce low-quality products to reduce costs, which makes it necessary for buyers to solve the "quality control" problem. In the meantime, sellers also face a reputation problem: If they once produce and sell a low-quality product, then their product will no longer be sold in the market in the next period, since their poor behavior will be exposed after buyers receive the goods. However, they can resell the asset to obtain a one-time benefit of v. The shipment requires f units of labor. Total cost, including production and delivery for the seller, is $x\theta + f$. Buyers possess all the bargaining power and can choose to integrate with the seller or not. If they choose to integrate, they will own the asset and the resale value v. If they do not integrate, they will only have the right to distribute the products produced by the sellers to the final consumers, and the resale value v is owned by the seller.

Under A style sourcing, buyers ensure quality by inspecting each shipment.[7] Buyers pay the inspection cost m_A and know the information of quality with certainty. To ensure the feasibility of the analysis, I make three assumptions: First, $\frac{f+\frac{\bar{\theta}}{\omega}x_A}{1-e^{-\frac{x_A}{q}}} \leq v$ so that buyers who choose A style sourcing do not need to pay any premium, otherwise the higher unit price plus the inspection costs required for each shipment will prevent any buyer from choosing A style sourcing; second, the value of a low-quality product is zero for both sellers and buyers; third, if buyers integrate sellers by owning the asset as well as the resale value v, reselling the asset is less profitable than selling a high-quality product to consumers. Eventually, in equilibrium, the reputation issues will cause producers that once delivered low-quality products permanently exit the market for this product. Based on these assumptions, the minimum wage that buyers are willing to pay for sellers to participate when choosing A style is

$$v_A(x_A, \bar{\theta}) = f + \bar{\theta}\omega x_A. \tag{2}$$

[7] Taylor and Wiggins (1997) allows buyers inspect with a probability $\bar{\alpha} = \frac{\bar{\theta}x - \underline{\theta}x}{f + \bar{\theta}x}$ instead of inspecting every shipment. As x_A is large, the asymptotic value of $\bar{\alpha}$ is $\frac{\bar{\theta}x - \underline{\theta}x}{\bar{\theta}x} = 1 - \frac{\underline{\theta}}{\bar{\theta}}$. When the gap between quality is large, the inspection probability goes to 1. Therefore, I assume that buyers who choose A style sourcing monitor every shipment with 100% probability. This is consistent with the special case of Taylor and Wiggins (1997) and makes subsequent analysis tractable.

The total cost, including inspection, for each shipment is

$$c_A(x_A, \bar{\theta}) = f + \bar{\theta} x_A + m_A. \tag{3}$$

If the buyer integrates the seller, then the resale value v and the usage right are owned by the buyer. Baker et al. (2002) pointed out that without inspection, sellers would predict that, since buyers own the assets and the right to use the final products, even if they produce high-quality products, buyers would only keep the products without paying, so sellers would only produce low-quality products. However, in my setting, buyers will inspect the quality of the products costly, which makes it unprofitable for sellers (buyers) to produce (get) low-quality products. Therefore, asset ownership does not change the behaviors of buyers and sellers compared with the original model in Heise et al. (2024), which means buyers will still pay as equation (2) shows, their cost is also equation (3), and sellers are incentivized by inspection to produce inputs with high quality.

It can be seen that when the total quantity q is fixed, the larger the quantity per shipment x, the smaller the number of shipments $\frac{q}{x}$, and the smaller the average cost in each cycle for buyers who choose A style sourcing. To simplify notation and calculation, I assume productivity ω is 1. Then equation (3) can be rewritten as

$$c_A(x_A, \bar{\theta}) = f + \bar{\theta} x_A + m_A. \tag{4}$$

Different from A style sourcing, J style sourcing ensures quality by providing incentives via long-term repeated contracts instead of inspecting each shipment. Therefore, if sellers provide a low-quality product to buyers, they can still gain payment $v_J(x_J, \bar{\theta})$ as well as the resale value v and break the relational contract. Reputation issues also result in a permanent exit from the market for this product for these sellers.

Assume that in order to maintain a relational contract, the buyer will pay the seller before receiving the goods and observing the quality of the goods.[8] Therefore, when r is

[8]Even if the buyer has all the bargaining power, in actual long-term international trade, it is rare for the buyer to receive the goods before paying. Without monitoring, the buyer provides incentives to the seller by paying in advance. At the same time, in a long-term relationship, whether the first payment occurs before or after the goods are received has little effect on the present value of long-term benefits. Therefore, whether J style purchases are paid before or after quality is observed does not affect the subsequent analysis. From a mathematical perspective, if the buyer pays after observing the quality of the product, the following conditions must be met:

$$\frac{v_J(x_J, \bar{\theta}) - f - \bar{\theta} x_J}{1 - e^{-r\frac{x_J}{q}}} \geq -f - \underline{\theta} x_J + v,$$

which means the payment $c_J(x_J, \bar{\theta}) = v_J(x_J, \bar{\theta}) = f + \bar{\theta} x_J + (1 - e^{-r\frac{x_J}{q}})(v - f - \underline{\theta} x)$. If $a \leq 0$, the maximum cost of J style is the same as the cost of A style when the monitoring cost m_A is 0, then J style will dominate A style, which means that all firms will choose J style, which is inconsistent with reality and is not the scope

the interest rate for each period and the discount is with continuous compounding, the per-shipment payment needs to satisfy the participant constraint and the incentive compatibility constraint:

$$v_J(x_J, \bar{\theta}) - f - \bar{\theta} x_J \geq 0, \tag{5}$$

and

$$\frac{v_J(x_J, \bar{\theta}) - f - \bar{\theta} x_J}{1 - e^{-r\frac{x_J}{q}}} \geq v_J(x_J, \bar{\theta}) - f - \underline{\theta} x_J + v. \tag{6}$$

Equation (5) ensures that the seller can at least get the payment to cover the cost of producing a high-quality product. The left-hand side of the inequality (6) is the net present value of the payment within the shipment cycle, and the right-hand side of it is the expected benefit of deviating from the contract and delivering a low-quality product. The resale value v does not appear on the left-hand side of inequality (6) because the specific asset will not be sold until the end of the relationship without breaking the contract, which means the present value of the asset from the future is expected to be zero. Inequality (6) ensures that revenue from a long-term relationship exceeds the one-time benefit of breaking the contract.

The optimal payment is given by taking equation (6) as binding. The right-hand side of inequality (6) must be greater than 0, otherwise the supplier will never deviate from a contract, which ensures the participant constraint (5) is satisfied. For the incentive compatibility constraint, the buyer can keep reducing the payment amount $v_J(x_J, \bar{\theta})$ until the left and right sides of equation (6) are equal. Any payment greater than the $v_J(x_J, \bar{\theta})$ that makes the left and right sides of equation (6) equal is not optimal for the buyer. When equation (6) is binding, the payment per shipment under J style sourcing is

$$c_J(x_J, \bar{\theta}) = v_J(x_J, \bar{\theta}) = f + \bar{\theta} x_J + (e^{r\frac{x}{q}} - 1)(\bar{\theta} - \underline{\theta}) x_J + (e^{r\frac{x}{q}} - 1)v. \tag{7}$$

Compared to equation (4), equation (7) omits the monitoring cost m_A but adds two additional premiums: the first is the quality-control premium $(e^{r\frac{x}{q}} - 1)(\bar{\theta} - \underline{\theta})x_J$. The second is the asset-control premium $(e^{r\frac{x}{q}} - 1)v$, which guarantees that suppliers will deliver goods rather than sell the assets if they own them. If the buyer chooses to integrate the seller, the seller will not have the choice to sell the asset, so the last term changes to zero. Buyers choose either the A style or the J style sourcing and maximize the expected profit:

$$\pi_s = \frac{\left[\int_0^{\frac{x_s}{q}} e^{-rt} pq \mathrm{d}t - c_s(x_s, \bar{\theta})\right]}{1 - e^{-r\frac{x_s}{q}}}, \ s \in \{J, A\}. \tag{8}$$

of interest of this paper. As long as $a > 0$, the subsequent analysis of A style vs J style will not be affected by whether the transaction is paid first or later, because there must be a critical value that makes firms choose different styles around this point.

4.2 Impact of Integration under Each Sourcing Style

Choosing the optimal quantity per shipment needs to solve the first-order condition:

$$\frac{c_s'(x_s,\bar{\theta})}{1-e^{-rx_s/q}} = \frac{c_s\frac{r}{q}e^{-rx_s/q}}{\left(1-e^{-rx_s/q}\right)^2},\ s\in\{J,A\}\,, \tag{9}$$

which means that the optimal x_s is such that the present value of the higher costs caused by the increase in order size (left-hand side) is equal to the cost savings from pushing these larger orders further into the future, thereby increasing the discount rate (right-hand side). The average cost is also the market price under each style:

$$p_s = AC_s(x_s,q) = \frac{r}{q}\frac{c_s}{1-e^{-rx_s/q}},\ s\in\{J,A\}\,. \tag{10}$$

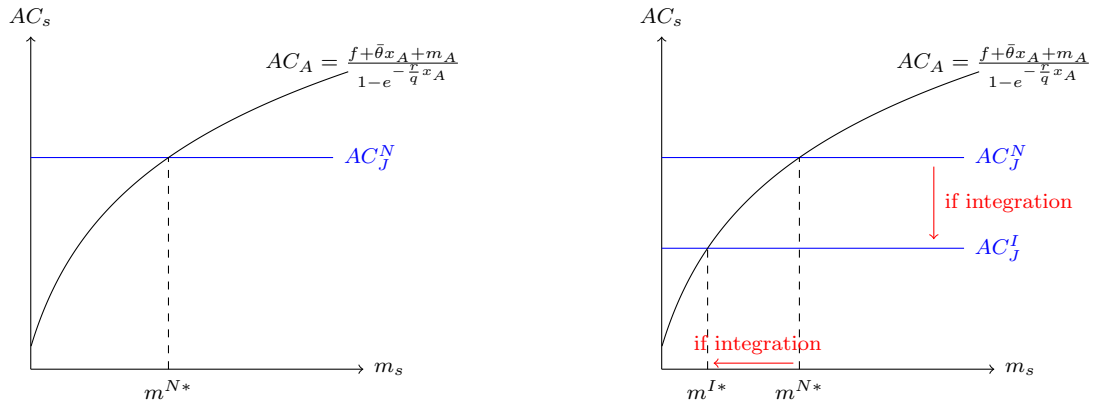
The only difference in my setting from theirs is that c_J includes a term $(e^{r\frac{x}{q}}-1)v$ when sellers own the asset.

When $m_A = 0$, the A style dominates the J style because the payment per shipment is lower for the A style when the inspection cost disappears. The average cost of the A style increases in m_A, while the average cost of the J style is independent of m_A. Thus, there is always a threshold value $m^* \in (0,\infty)$ such that the average cost under both styles is the same, as Figure 2a shows. Buyers will choose the A style when $m < m^*$ and choose the J style when $m > m^*$.

When the buyer integrates the seller, the last term in (7) vanishes to zero, and the average cost of J style decreases. As Figure 2b shows that integration results in AC_J^N moves to AC_J^I, which makes the critical value of m changes from m^{N*} to m^{I*}. Since the firm will choose the J style when $m > m^*$, a decrease in m^* makes it easier for the firm to choose the J style, which means that integration makes it easier for the firm to switch from the A style to the J style. Owning assets allows sellers to get both a period of payment and a profit from the resale of assets before breaking the relational contract. They are more likely to choose to produce a low-quality product. Buyers need to spend more to give sellers incentives to obtain a high-quality product. Integration allows buyers to have the right to use and sell assets, and the benefits that sellers can get from breach of contract are reduced. Compared with the case of non-integration, J style is easier to ensure product quality after integration, so integration and J style are complementary.

Proposition 1 *Vertical integrations between downstream firms and upstream firms make it easier for buyers to choose Japanese style sourcing. The average quantity per shipment x in the market will decrease.*

Figure 2: The Change of Average Cost in m



(a) Not Integration (b) Integration

Notes. The figure shows the change in the average cost under J style and the change in the m^* after the downstream firm integrates the upstream firm. When $m > m^{N*}$, integrations enable buyers to maintain the option of choosing J style sourcing. When $m < m^{I*}$, integrations enable buyers to maintain the option of choosing A style sourcing. Only when $m^{I*} < m < m^{N*}$, integrations will cause buyers who originally chose A style sourcing to switch to J style sourcing. Therefore, I need to analyze the changes in x_s under all three cases in order to prove that changes in the average x_s of the whole market decrease after integrations.

In the empirical analysis, I use the size of quantity per shipment x_s to distinguish J style from A style, and J style has a smaller x_s. Therefore, I am going to prove that when v decreases to 0, the average x_s decreases.

Proof. See Appendix A.1. ■

4.3 Comparative Statics and Additional Predictions

From the analysis of obtaining the main prediction, it can be seen that the main factors affecting the downstream firms' choice between A style procurement and J style procurement are the size of the inspection cost m_A and the strength of the upstream firms' willingness to produce low-quality products. This section focuses on discussing whether the complementarity between J style sourcing and integration is strong or weak in the face of different factors affecting upstream firms' default possibilities.

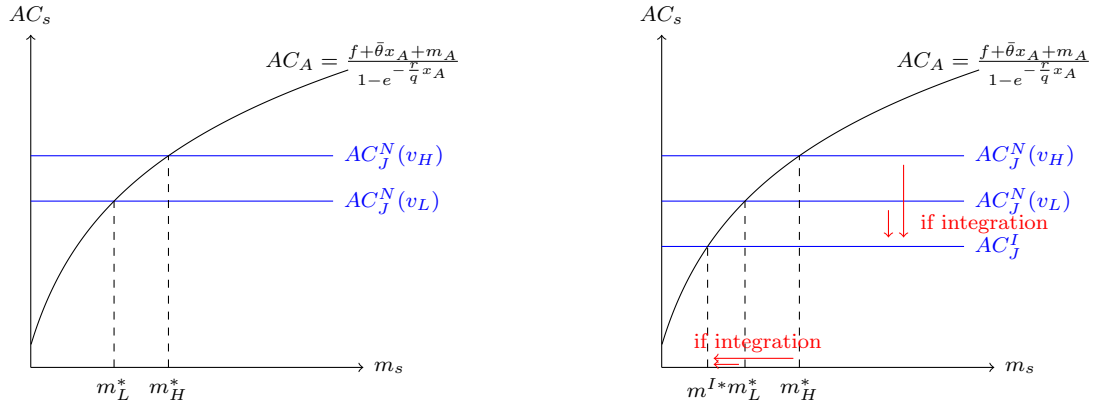
4.3.1 The Impact of Resale Value

The size of v that can be obtained from selling assets directly affects the one-time benefits that the seller can get after producing low-quality products. Therefore, the size of v will affect the strength of complementarity between J style sourcing and integration. Intuitively, with other factors remaining unchanged, the larger v is, the more likely the seller is to produce low-quality products and terminate the contract to exit the market. Therefore, on the one

hand, integration is more likely to influence sellers' decisions on whether to deviate from existing relationships, and on the other hand, it can make buyers spend less when choosing J style sourcing. Therefore

Proposition 2 *When the resale value of asset v is larger, the complementarity between Japanese style sourcing and integration should be stronger. Integration leads more firms to choose to switch from A style to J style, which will reduce the average quantity per shipment x in the market.*

Figure 3: The Change of Average Cost for Different v



(a) Not Integration (b) Integration

Notes. The figure shows the change in the average cost under J style and the change in the m^* after the downstream firm integrates the upstream firm under different sizes of the resale value v_H and v_L.

Figure 3 shows the change of average cost in m for different sizes of resale value v. When other variables are fixed, and there is no integration, the only difference brought by different sizes of v is the average cost of J style sourcing. The larger v is, the greater the average cost of J style sourcing is, and thus the greater m^* is. Integration reduces the average cost of the two cases to the same level, and both m_L^* and m_H^* are also reduced to m^{I*}. When buyers face sellers with v_H and v_L respectively, different levels of m^* divide all possible distributions of m into four parts: Firstly, when $m > m_H^*$, the average cost of A style sourcing is higher because the inspection cost is too high. If there is no integration, buyers will choose to use J style. If buyers acquire sellers, the average cost of J style will drop further, causing them to maintain J style unchanged. Secondly, when $m_L^* < m \leq m_H^*$, if there is no integration, choosing to maintain J style procurement from sellers with v_H will face greater average costs, while choosing to maintain J style procurement from sellers with v_L is still a more economical option. Therefore, the integration will change the transaction style between buyers and sellers with v_H from A to J, while the transaction style between

buyers and sellers with v_L will remain J. Thirdly, when $m^{I*} < m \leq m_L^*$, inspection costs are further reduced, which makes A style procurement a better choice regardless of which seller to import from when there is no integration, while integration will cause buyers to choose J style procurement instead. Finally, when $m \leq m^{I*}$, since the monitoring costs are very low, buyers will maintain A style purchasing regardless of whether they are trading with v_H or v_L, and regardless of whether an integration occurs.

As Proof 4.2 mentioned, only the firms that choose to switch from A style to J style will significantly affect the average x in the market. Based on Figure 3 and the discussions above, buyers will switch from the A style to the J style when the inspection cost m_A satisfies $m^{I*} < m_A \leq m_H^*$ when integrating a supplier that has a large value of asset. However, if they integrate a supplier that has a lower value of assets, they switch only when $m^{I*} < m \leq m_L^*$. Therefore, as long as $m_A \in [0, \infty)$ follows any positive random distribution, buyers who integrate sellers with higher asset values have a greater probability of switching from A style to J style. Eventually, the higher the value of the assets, the stronger the complementarity between vertical integration and the J style sourcing, and also the smaller the quantity per shipment x. The size of the asset value has a different impact because that integration reduces the value of outside options more for suppliers with larger asset values.

5 Additional Empirical Analysis

This section describes the empirical analysis for testing the additional predictions. I interact with some supplier characteristics and the indicator of whether the buyer and supplier are related parties to examine the heterogeneous impact of different types of firms. Similar to the motivating fact, the results in this section only show firm-level relationships rather than causality. Although this will have endogeneity problems, I show the robustness of the results by flexibly controlling for different fixed effects.

5.1 Empirical Results of Comparative Statics

First, I show that the strength of complementarity between a firm pair's integration status and its J style sourcing varies when subsidiaries have different asset sizes.

Table 2 shows the results of the impact of heterogeneous subsidiary size on the main effect. I use an indicator variable that equals 1 when trading volume $\times$ ownership share exceeds the 75% percentile in the current year as a proxy for large subsidiaries, because asset size is typically very large and highly variable. The first four columns show that, given that related parties are more likely to choose J style sourcing, if the assets of the subsidiary controlled by the owner are larger, the frequency of shipping will be further increased by

20%-40% compared with the partner with smaller assets of the subsidiary controlled by the owner.

Table 2: Heterogeneous Effect, using the Indicator of Large Subsidiary

Dep. Var.	$\log(Avg\ \#\ Shipment_{ijt})$				$\log(Avg\ \frac{quantity_{ijt}}{shipment_{ijt}})$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\}$	0.167^a	0.066^a	0.265^a	0.156^a	-0.172^a	-0.073^a	-0.273^a	-0.166^a
	(0.014)	(0.014)	(0.043)	(0.045)	(0.014)	(0.014)	(0.043)	(0.045)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\} \times \mathbb{1}\{\text{Large Subsidiary}\}$		0.313^a		0.178^b		-0.304^a		-0.175^b
		(0.030)		(0.069)		(0.030)		(0.070)
$\log(TotQuantity_{ijt})$	0.461^a	0.461^a	0.511^a	0.510^a	0.536^a	0.536^a	0.488^a	0.488^a
	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)
Importer FE	Y	Y	N	N	Y	Y	N	N
Exporter FE	Y	Y	N	N	Y	Y	N	N
Year FE	Y	Y	N	N	Y	Y	N	N
1st Transaction Mon. FE	Y	Y	N	N	Y	Y	N	N
Importer-Year FE	N	N	Y	Y	N	N	Y	Y
Exporter Year FE	N	N	Y	Y	N	N	Y	Y
Importer-1st Transaction Mon. FE	N	N	Y	Y	N	N	Y	Y
Exporter-1st Transaction Mon. FE	N	N	Y	Y	N	N	Y	Y
Observations	5553711	5553711	2512432	2512432	5553711	5553711	2512432	2512432
# of Firm Pairs	2905188	2905188	1358234	1358234	2905188	2905188	1358234	1358234
R-squared	0.825	0.825	0.900	0.900	0.928	0.928	0.950	0.950

Notes: Standard errors are clustered at the firm-pair level. A "large subsidiary" is defined as a subsidiary with a trading volume × ownership share greater than 75 percentile in the current year. There are 679 pairs of firms that are identified as related parties by the csh file without any information about the ownership share percentage in the ORBIS data. They account for approximately 13.86% of all related parties. In order to prevent these firms from influencing my analysis, when verifying Proposition 2, I directly remove these firms from my sample; that is, I neither consider them to be related parties nor to be independent firms. The ownership share refers total ownership share. The results are robust when using the direct ownership share. The total number of observations is smaller than that in Table 1 because there are some firms that are identified as related parties by csh files but do not have share information, so I drop these pairs from the sample. Significance levels: a: 0.01, b: 0.05, c: 0.10.

5.2 Additional Evidence on Long-term Relationship of J Style Sourcing

My model only predicts how the quantity per shipment will respond to a firm's vertical integration. In this section, I will present some facts that are not predicted by the model but can support the idea that related parties are more likely to choose the J style. Based on the assumptions of A style and J style, buyers who choose A style typically use competitive bidding to select the lowest-cost supplier, while those who choose J style typically renew their contracts with the same supplier. We can assume that, on average, buyers who choose A style have no reason to do business with the same supplier for a long time, while buyers who choose J style are more likely to stick with the same firm. Table 3 shows that, given the same

firm-pair, being a related party is more likely to have more frequent transactions with the same supplier in the transaction cycle, that is, the average time between shipments is shorter. More specifically, if the time difference between the first and last transaction between two firms is the same, then A style is more likely to randomly make several transactions with the same supplier, and J style is more likely to frequently renew contracts with the same supplier during this time. When the two firms have a vertical integration relationship, their trading pattern is closer to J style-they have shorter average months between shipments.

Table 3: Effect on Average Time between Shipments

Dep. Var.	Log Avg Month between Shipments			
	(1)	(2)	(3)	(4)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\}$	-0.107^a	-0.096^a	-0.108^a	-0.270^a
	(0.012)	(0.013)	(0.012)	(0.043)
$\log(TotQuantity_{ijt})$	-0.316^a	-0.337^a	-0.317^a	-0.373^a
	(0.001)	(0.001)	(0.001)	(0.001)
Importer FE	Y	N	Y	N
Exporter FE	Y	N	Y	N
Year FE	N	N	Y	N
1st Transaction Mon. FE	Y	N	Y	N
Importer-Year FE	N	N	N	Y
Exporter-Year FE	N	N	N	Y
Importer-1st Transaction Mon. FE	N	Y	N	Y
Exporter-1st Transaction Mon. FE	N	Y	N	Y
Observations	3347338	2502449	3347338	1394906
# of Firm Pairs	1401997	1047802	1401997	661617
R-squared	0.697	0.766	0.698	0.807

Notes: Standard errors are clustered at the firm-pair level. The same moving average approach is used in these regressions. The average month between shipments is calculated by $\frac{\text{last date}_{ijt} - \text{first date}_{ijt}}{30}$. I also control for year fixed effects in the empirical models to absorb the time-varying factors. Significance levels: a: 0.01, b: 0.05, c: 0.10.

In addition, importers who choose J style will also have fewer suppliers per shipment and longer average relationship length with suppliers. Borrowing Heise et al. (2024)'s definition of supplier per shipment and average length aggregated at the buyer level, I construct similar variables and find that when a buyer has a higher proportion of relational suppliers, its supplier per shipment is smaller and its average length with suppliers is longer, which is consistent with the characteristics of J style. The results are shown in Table 4.

Table 4: Effect on Supplier per Shipment and Average Length

Dep. Var.	Log Supplier per Shipment		Log Avg Length	
	(1)	(2)	(3)	(4)
$\frac{\#\text{ Related Supplier}_{it}}{\#\text{ Supplier}_{it}}$	-0.150^a	-0.185^a	0.170^a	0.217^a
	(0.016)	(0.020)	(0.021)	(0.023)
$\log(TotQuantity_{it})$	-0.256^a	-0.307^a	0.163^a	0.193^a
	(0.001)	(0.001)	(0.002)	(0.002)
Importer FE	Y	N	Y	N
Year FE	Y	Y	Y	Y
1st Transaction Mon. FE	Y	N	Y	N
Importer-1st Transaction Mon. FE	N	Y	N	Y
Observations	1418882	687705	840751	525931
# of Firm Pairs	360404	183686	203375	130955
R-squared	0.834	0.881	0.586	0.631

Notes: Standard errors are clustered at the importer level. The same moving average approach is used in these regressions. The supplier per shipment is the ratio of the number of suppliers to the number of shipments, which has the largest value of 1 (every shipment has a different supplier). The average length of all the suppliers is calculated by $\frac{\sum_j \text{Months between Shipments}_{jt}}{\#\text{Suppliers}_{it}}$. I also control for year fixed effects in the empirical models to absorb the time-varying factors. Significance levels: a: 0.01, b: 0.05, c: 0.10.

6 Conclusions

This paper studies the relationship between vertical integration and firms' choice of sourcing styles. I analyze the relational contract within business groups both theoretically and empirically. I build a theoretical model to show that when firms are vertically integrated, they are more likely to choose J style sourcing, which is characterized by high frequency and low volume of shipments. The model also shows that the complementarity between vertical integration and J style sourcing is stronger when the subsidiary controlled by the parent firm has larger asset values or when the quality gap of the product traded is higher. I use firm-pair level trade data combined with ownership data to verify the theoretical predictions. The empirical results show that related parties have more shipments and less quantity per shipment than independent firms, which is consistent with the theoretical model. In addition, the empirical results also support the comparative statics of the theoretical model. I also provide some supportive facts that transactions between related parties are more likely to be align with the characteristics of J style sourcing, such as shorter average time between shipments, fewer suppliers per shipment for each importer, and longer average relationship length of suppliers.

Although I cannot use exogenous variation to identify the causal relationship between related parties and J style sourcing, my theoretical model and empirical results show that the relationship between the two is robust. My research shows that the form of firm and the

procurement strategy interact with each other—firms can control their procurement costs through various means and choose different sourcing systems based on the characteristics of transaction partners and the features of the traded products.

References

Alfaro, Laura, Pol Antràs, Davin Chor, and Paola Conconi, "Internalizing Global Value Chains: A Firm-Level Analysis," *Journal of Political Economy*, April 2019, *127* (2), 508–559.

Antràs, Pol and Davin Chor, "Organizing the Global Value Chain," *Econometrica*, 2013, *81* (6), 2127–2204. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA10813.

__ **and Elhanan Helpman**, "Contractual Frictions and Global Sourcing," Working Paper 12747, National Bureau of Economic Research December 2006. Series: Working Paper Series.

__ **, Evgenii Fadeev, Teresa C Fort, and Felix Tintelnot**, "Global Sourcing and Multinational Activity: A Unified Approach," Working Paper 30450, National Bureau of Economic Research September 2022. Series: Working Paper Series.

Baker, George, Robert Gibbons, and Kevin J. Murphy, "Relational Contracts and the Theory of the Firm*," *The Quarterly Journal of Economics*, February 2002, *117* (1), 39–84. _eprint: https://academic.oup.com/qje/article-pdf/117/1/39/5238371/117-1-39.pdf.

Berlingieri, Giuseppe, Frank Pisch, and Claudia Steinwender, "Organizing Global Supply Chains: Input-Output Linkages and Vertical Integration," *Journal of the European Economic Association*, June 2021, *19* (3), 1816–1852.

Costinot, Arnaud, Lindsay Oldenski, and James Rauch, "ADAPTATION AND THE BOUNDARY OF MULTINATIONAL FIRMS," *The review of economics and statistics*, 2011, *93* (1), 298–308. Place: Cambridge Publisher: MIT Press.

Fernandes, Ana and Heiwai Tang, "Fast Fashion: Theory and Evidence from Portuguese Textile and Clothing Firms," *SSRN Electronic Journal*, 2020.

Forbes, Silke Januszewski and Mara Lederman, "Adaptation and Vertical Integration in the Airline Industry," *American Economic Review*, December 2009, *99* (5), 1831–1849.

Fort, Teresa C., "The Changing Firm and Country Boundaries of US Manufacturers in Global Value Chains," *Journal of Economic Perspectives*, 2023, *37* (3), 31–58.

Gibbons, Robert, "Four formal(izable) theories of the firm?," *Journal of Economic Behavior & Organization*, October 2005, *58* (2), 200–245.

Grossman, Gene M. and Elhanan Helpman, "Managerial incentives and the international organization of production," *Journal of International Economics*, 2004, *63* (2), 237–262.

Heise, Sebastian, Justin R. Pierce, Georg Schaur, and Peter K. Schott, "Tariff Rate Uncertainty and the Structure of Supply Chains," *International Finance Discussion Paper*, April 2024, (1389), 1–92.

Kalemli-Ozcan, Sebnem, Bent E. SSrensen, Carolina Villegas-Sanchez, Vadym Volosovych, and Sevcan Yesiltas, "How to Construct Nationally Representative Firm Level Data from the ORBIS Global Database," *SSRN Electronic Journal*, 2015.

Milgrom, Paul and John Roberts, "The Economics of Modern Manufacturing: Technology, Strategy, and Organization," *The American economic review*, 1990, *80* (3), 511–528. Place: Menasha, Wis Publisher: American Economic Association.

Ortiz, Julio L., "Spread Too Thin: The Impact of Lean Inventories," *International Finance Discussion Papers*, April 2022, *2022.0* (1342), 1–74. Publisher: Board of Governors of the Federal Reserve System.

Taylor, Curtis and Steven N Wiggins, "Competition or Compensation: Supplier Incentives under the American and Japanese Subcontracting Systems," *American Economic Review*, 1997, *87* (4), 598–618.

Wasi, Nada and Aaron Flaaen, "Record Linkage Using Stata: Preprocessing, Linking, and Reviewing Utilities," *The Stata Journal: Promoting communications on statistics and Stata*, October 2015, *15* (3), 672–697.

Online Appendix

A Proof of Model Predictions

A.1 Proof of Proposition 1

If buyers keep choosing A style sourcing, x_A will not change regardless of integration. However, if buyers keep choosing J style sourcing, integration will change x_J:

$$\begin{aligned}\frac{\partial FOC_J}{\partial v} &= (1-e^{-r\frac{x_J}{q}})\frac{r}{q}e^{r\frac{x_J}{q}} - (e^{r\frac{x_J}{q}}-1)\frac{r}{q}e^{-r\frac{x_J}{q}} \\ &= \frac{r}{q}(e^{r\frac{x_J}{q}}+e^{-r\frac{x_J}{q}}-2)\end{aligned} \tag{A-1}$$

Because $0 < r\frac{x_J}{q} < 1$, $e^{r\frac{x_J}{q}}$ and $e^{-r\frac{x_J}{q}}$ must be great than 0. So $e^{r\frac{x_J}{q}}+e^{-r\frac{x_J}{q}} > 2$, which implies that $\frac{\partial FOC_J}{\partial v} > 0$

$$\begin{aligned}SOC_J = \left(1-e^{-\frac{rx_J}{q}}\right)\left(2\frac{r}{q}\frac{\bar{\theta}-\underline{\theta}}{\omega}e^{\frac{rx_J}{q}} + \frac{r^2}{q^2}ve^{\frac{rx_J}{q}} + \frac{r^2}{q^2}\frac{x_J(\bar{\theta}-\underline{\theta})}{\omega}e^{\frac{rx_J}{q}}\right) \\ + \frac{r^2}{q^2}\left(f + v\left(e^{\frac{rx_J}{q}}-1\right) + \frac{\bar{\theta}x_J}{\omega} + \frac{x_J(\bar{\theta}-\underline{\theta})\left(e^{\frac{rx_J}{q}}-1\right)}{\omega}\right)e^{-\frac{rx_J}{q}}\end{aligned} \tag{A-2}$$

Both terms of (A-2) are larger than 0 because $1-e^{-r\frac{x_J}{q}} > 0$ ($e^{\frac{rx_J}{q}}-1>0$) and $\bar{\theta}-\underline{\theta} > 0$. The summation $SOC_J > 0$. Apply the implicit function theorem:

$$\frac{dx_J^*}{dv} = \frac{-\frac{\partial FOC_J}{\partial v}}{SOC_J} < 0,$$

which means as v decreases to 0, x_J increases $\frac{\frac{\partial FOC_J}{\partial v}}{SOC_J}$ units. But in reality, the interest rate r is small, and the $\frac{x_J}{q}$ of J style purchases is also small. When there is a small r and a large q, the increase in x_J is close to 0.

If buyers switch from A style to J style, the quantity per shipment x_s will decrease greatly because x_A must be greater than x_J, which is proved in Taylor and Wiggins (1997)'s Proposition 3 and is also one of the biggest differences between A style and J style mentioned in this study.

In general, if m is uniformly distributed, then firms that switch from A style to J style will increase the number of firms using J style in the market and reduce the average x_s.

B Data Refinement

B.1 Identifying MNEs using ORBIS Data

According to the mutual ownership relationship between firms, the magnitude of controlling shareholder levels increases successively (firms with large csh levels own firms with small csh levels). I regard the firms with the largest csh levels as the ultimate owners. Therefore, in an "ownership tree", if the firm with the largest csh level is a non-US firm, then all the US firms within the tree are defined as foreign MNEs. If the firm with the largest csh level is a US firm, then as long as there is a non-US firm within the tree with a smaller csh level, this US firm that has the largest csh level is defined as a US MNE. At the same time, if there is at least one non-US firm with a smaller csh level than other US firms in the tree, then these US firms with relative larger csh levels will also be regarded as US MNEs.[9] Because I want to study the global sourcing behavior of US firms, this approach excludes business groups that are either entirely non-US firms or entirely US domestic firms.

B.2 Details on Matching Procedure

Two challenges need to be addressed when matching firms in two datasets by name and country: First, some studies choose to use fuzzy matching to keep observations with higher matching scores, but fuzzy matching cannot guarantee the quality of the matching. Second, the formats of firm names in different datasets are different. Even the same firm may have different spellings in the same dataset. If I want to match firm names accurately instead of using fuzzy matching, it is necessary to process the firm names to find as many firms as possible that have different spellings of their names but are actually the same firm.

Based on the pattern files provided by Wasi and Flaaen (2015), which includes common symbols, words and their corresponding abbreviations in firm names, possible forms of numbers, etc., as well as the regularities of firm names in these two datasets that I observed, I uniformly replaced and deleted certain elements, and then removed suffixes such as CO., LTD. to obtain a clean base name. Eventually, I match two datasets using the clean names and country code.

[9] For example, if a tree is arranged from high to low in terms of csh level as GBR1-USA1-FRA1-USA2, then USA1 and USA2 are both foreign MNEs. If it is arranged as USA1-GBR1-FRA1-USA2, then USA1 is a US MNE. If it is arranged as USA1-GBR1-USA2-FRA1, then USA1 and USA2 are both US MNEs.

C Additional Empirical Results

C.1 Robustness Check of Main Results

Table C-1: Relationship between Shipment Pattern and Related Party, Excluding Japanese Firms

Dep. Var.	$\log(Avg\ \#\ Shipment_{ijt})$		$\log(Avg\ \frac{quantity_{ijt}}{shipment_{ijt}})$		$\log(\frac{Avg\ quantity_{ijt}}{Avg\ shipment_{ijt}})$	
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\}$	0.157^a	0.234^a	-0.160^a	-0.242^a	-0.157^a	-0.234^a
	(0.012)	(0.038)	(0.012)	(0.038)	(0.012)	(0.038)
$\log(TotQuantity_{ijt})$	0.462^a	0.512^a	0.535^a	0.486^a	0.538^a	0.488^a
	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
Importer FE	Y	N	Y	N	Y	N
Exporter FE	Y	N	Y	N	Y	N
Year FE	Y	N	Y	N	Y	N
1st Transaction Mon. FE	Y	N	Y	N	Y	N
Importer-Year FE	N	Y	N	Y	N	Y
Exporter Year FE	N	Y	N	Y	N	Y
Importer-1st Transaction Mon. FE	N	Y	N	Y	N	Y
Exporter-1st Transaction Mon. FE	N	Y	N	Y	N	Y
Observations	5466757	2479894	5466757	2479894	5466757	2479894
# of Firm Pairs	2865368	1341896	2865368	1341896	2865368	1341896
R-squared	0.825	0.900	0.928	0.950	0.929	0.951

Notes: Standard errors are clustered at the firm-pair level. Moving averages are use for dependent variables and control variable $\log(TotQuantity_{ijt})$. The dependent variables in columns (3)-(6) are all quantity per shipment, but the third and fourth columns directly take the moving average of quantity per shipment in the corresponding years, while the fifth and sixth columns use the moving average of quantity and shipment respectively to calculate the quantity per shipment. The results of the two methods of calculating quantity per shipment are consistent with each other. Significance levels: a: 0.01, b: 0.05, c: 0.10.

C.2 Robustness Check of Additional Predictions

Table C-2: Heterogeneous Effect, using the Indicator of Large Subsidiary, Direct Ownership Share

Dep. Var.	$\log(Avg\ \#\ Shipment_{ijt})$				$\log(Avg\ \frac{quantity_{ijt}}{shipment_{ijt}})$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\}$	0.168^a	0.061^a	0.287^a	0.092^b	-0.170^a	-0.065^a	-0.287^a	-0.089^b
	(0.010)	(0.010)	(0.033)	(0.037)	(0.011)	(0.010)	(0.033)	(0.037)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\} \times \mathbb{1}\{\text{Large Subsidiary}\}$		0.346^a		0.330^a		-0.339^a		-0.335^a
		(0.023)		(0.055)		(0.023)		(0.055)
$\log(TotQuantity_{ijt})$	0.461^a	0.461^a	0.511^a	0.511^a	0.536^a	0.536^a	0.488^a	0.488^a
	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)
Importer FE	Y	Y	N	N	Y	Y	N	N
Exporter FE	Y	Y	N	N	Y	Y	N	N
Year FE	Y	Y	N	N	Y	Y	N	N
1st Transaction Mon. FE	Y	Y	N	N	Y	Y	N	N
Importer-Year FE	N	N	Y	Y	N	N	Y	Y
Exporter Year FE	N	N	Y	Y	N	N	Y	Y
Importer-1st Transaction Mon. FE	N	N	Y	Y	N	N	Y	Y
Exporter-1st Transaction Mon. FE	N	N	Y	Y	N	N	Y	Y
Observations	5559808	5559808	2515322	2515322	5559808	5559808	2515322	2515322
# of Firm Pairs	2906008	2906008	1359309	1359309	2906008	2906008	1359309	1359309
R-squared	0.826	0.826	0.900	0.900	0.928	0.928	0.950	0.950

Notes: Standard errors are clustered at the firm-pair level. A "large subsidiary" is defined as a subsidiary with a trading volume × ownership share greater than 75 percentile in the current year. The ownership share refers direct ownership share. The results are robust when using direct ownership share. The total number of observations is smaller than that in Table 1 because there are some firms that are identified as related party by csh files but do not have share information so that I drop these pairs from the sample. Significance levels: a: 0.01, b: 0.05, c: 0.10.

Table C-3: Heterogeneous Effect, using the Indicator of Medium Subsidiary

Dep. Var.	$\log(Avg\ \#\ Shipment_{ijt})$				$\log(Avg\ \frac{quantity_{ijt}}{shipment_{ijt}})$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\}$	0.168^a	0.006	0.287^a	-0.004	-0.170^a	-0.012	-0.287^a	0.009
	(0.010)	(0.011)	(0.033)	(0.054)	(0.011)	(0.011)	(0.033)	(0.054)
$\mathbb{1}\{\text{Related Party Trade}_{ijt}\} \times \mathbb{1}\{\text{Medium Subsidiary}\}$		0.278^a		0.342^a		-0.270^a		-0.348^a
		(0.017)		(0.062)		(0.017)		(0.062)
$\log(TotQuantity_{ijt})$	0.461^a	0.461^a	0.511^a	0.511^a	0.536^a	0.536^a	0.488^a	0.488^a
	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)
Importer FE	Y	Y	N	N	Y	Y	N	N
Exporter FE	Y	Y	N	N	Y	Y	N	N
Year FE	Y	Y	N	N	Y	Y	N	N
1st Transaction Mon. FE	Y	Y	N	N	Y	Y	N	N
Importer-Year FE	N	N	Y	Y	N	N	Y	Y
Exporter Year FE	N	N	Y	Y	N	N	Y	Y
Importer-1st Transaction Mon. FE	N	N	Y	Y	N	N	Y	Y
Exporter-1st Transaction Mon. FE	N	N	Y	Y	N	N	Y	Y
Observations	5559808	5559808	2515322	2515322	5559808	5559808	2515322	2515322
# of Firm Pairs	2906008	2906008	1359309	1359309	2906008	2906008	1359309	1359309
R-squared	0.826	0.826	0.900	0.900	0.928	0.928	0.950	0.950

Notes: Standard errors are clustered at the firm-pair level. A "large subsidiary" is defined as a subsidiary with a trading volume $\times$ ownership share greater than 50 percentile in the current year. The ownership share refers direct ownership share. The results are robust when using direct ownership share. The total number of observations is smaller than that in Table 1 because there are some firms that are identified as related party by csh files but do not have share information so that I drop these pairs from the sample. Significance levels: a: 0.01, b: 0.05, c: 0.10.